Code No: 124CB **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, November/December - 2020
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the properties of algorithm with real time example.
b) How was the Strassen's matrix multiplication better than general matrix multiplication method? [7+8]

2.a) Explain non recursive in order tree traversal technique with example.
b) List the differences between connected and bi-connected components. [7+8]

3. Write an algorithm of OBST. [15]

4. Write a back tracking algorithm for sum of subset problem using the state space tree corresponding to the variable tuple size formulation. [15]

5.a) Write the differences between Deterministic polynomial, Non deterministic polynomial algorithms.
b) Explain about cook's theorem. [8+7]

6. Sort the following data using quick sort.
29, 14, 19, 10, 7, 17, 16, 4, 3, 5. [15]

7.a) Write an algorithm of weighted union.
b) Differentiate between BFS and DFS. [8+7]

8.a) Write an algorithm of Greedy knapsack.
b) Explain about the Reliability design. [8+7]

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